

Total number of printed pages-4

4R/2025 PHY4104C

2025

PHYSICS

(Major-I)

Paper : PHY4104C

(Electricity and Magnetism)

Full Marks : 45

Time : Two hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions in very short :

1×4=4

- (a) What will be the total electric flux, leaving a spherical surface of radius r and surrounding an electric dipole?
- (b) What is the capacitance of isolated charged sphere of radius R ?
- (c) Find the exponential form of the complex number j .

- (d) What is the phase difference between the current and the applied voltage in a series LCR (A.C.) circuit at resonance?

2. Answer the following questions in short:

$$2 \times 3 = 6$$

- (a) What is the ideal value of dielectric constant for conducting material? Justify your answer.
- (b) Show that stored energy in a capacitor is $\frac{1}{2} CV^2$.
- (c) Give the idea of displacement current in Electromagnetic theory.

3. Answer the following questions: $5 \times 3 = 15$

- (a) What are electric capacitors? Find the capacitance of spherical conductor enclosed by an earthen concentric spherical shell. 5

- (b) What are Polarization and polarization vector? Show that the magnitude of polarization vector is equal to polarization surface charge density.

$$3 + 2 = 5$$

Or

What do you mean by bound/polarization charge? Find the magnitude of polarization charge. 5

- (c) Deduce an expression for self-inductance of a long solenoid carrying current. 5

Or

An emf $e = E_m \sin \omega t$ is applied to a series LCR circuit. Obtain expression for instantaneous current using complex notations. What is the condition for resonance?

4. Answer **either** (a) **or** (b) : 10

(a) What are Poisson's and Laplace's equations? Applying Laplace's equation, find the potential and field between two concentric spherical conductors. Also find the potential and field between two concentric cylinders.

(b) Draw an illustrative diagram for the relation $D = \epsilon_0 E + P$ and explain. (Symbols have their usual meaning). Calculate the capacitance of a capacitor filled with two dielectrics of same dimension but of dielectric constants k_1 and k_2 respectively.

5. Answer **either** (a) **or** (b) :

10

- (a) State and explain the Biot-Savart law. Using Biot-Savart law, find the Magnetic field at a distance x from the centre, on the axis of a circular coil of radius R carrying current I .

A circular coil with 40 turns of radius 5×10^{-2} m is carrying a current 0.2 A. Find magnetic field at the centre of the coil.

- (b) Find the divergence of Magnetic Field $\vec{B}$. What is the physical significance of the result? State and prove Ampere's circuital law. Hence find the curl of Magnetic Field $\vec{B}$.

$4\pi \times 10^{-7}$